ELECTRIC VEHICLE MANAGEMMENT SYSTEM USING ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

COMMON INTEGRATED SOLUTION REPORT

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DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

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ABSTRACT

As Electric Vehicles (EVs) are becoming more prevalent, there is a growing need for reliable vehicle monitoring, battery management, charging support, and accessible technical information. But most of the currently available EV applications offer these services separately, and users end up having to use more than one service without a single system with integrated predictive and personalized decision support. To solve these problems, this research introduces IntelligentEV, a mobile decision support system based on Artificial Intelligence (AI) and Machine Learning (ML) to tackle various aspects of the problem in four inter-related components.

The first module, AI-Based EV Operational Anomaly Detection, uses the battery voltage, current, temp, motor, speed, acceleration, charging state and battery usage to detect abnormal operating conditions and offer diagnosis information. The second part is Adaptive Battery Health Prediction and Personalized Recommendation, which leverages past degradation information of lithium-ion batteries and supervised regression models such as Random Forest and XGBoost to predict the battery State of Health (SOH) after 3, 6 and 12 months. The user-specific charging and driving patterns are analyzed by a separate rule-based mechanism and used to provide personalized battery maintenance suggestions. The third part is Intelligent EV Charging Optimization and Range Prediction, which is used to predict the driving range of the battery, the available charging stations, the waiting line at charging stations, route planning, selecting charging stations, and predicting charging time and cost. The fourth feature is an AI-Based EV Virtual Assistant, which utilizes retrieval-augmented generation (RAG), large language models (LLMs), and text and voice interaction to provide context-relevant charging, operation, and maintenance guidance.

They are intended to be integral to an overall mobile application that includes vehicle diagnostics, battery forecasting, mobility planning and conversational assistance. It is designed to make it easier for EV owners to understand what's going on in their vehicle and to make better decisions for their daily and long-term driving. It involves modular integration, user-friendly guidance, and data-informed decision-making while acknowledging the significance of data privacy, quality, and reliability. It is important to further evaluate with a variety of real-world EV data and users for long-term applicability to various vehicles operating in various environments and charging conditions.

Keywords: Electric Vehicles, Artificial Intelligence, Machine Learning, Battery State of Health, Intelligent EV Management

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TABLE OF CONTENTS

1 INTRODUCTION AND PROJECT CONTEXT 10
1.1 Background and context 10
1.2 Stakeholders and user needs 12
1.2 Alignment with the Sustainable Development Goals 15
1.3 Purpose and scope of the common report 17
2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS 21
2.1 Overall problem definition 21
2.2 Root causes and overall research gap 22
2.2.1 Root causes of the problem 22
2.2.2 Overall research gap 23
2.3 Subproblem and component mapping 24
2.4 Overall aim and group objectives 27
2.4.2 Group objectives 27
2.4.3 Evaluation strategy and success criteria 30
2.5 Scope assumptions and constraints 31
2.5.1 Scope of the complete solution 31
2.5.2 Scope assumptions 32
2.5.3 Technical and operational constraints 33
2.5.4 Exclusions and boundaries 35
2.5.5 Dependencies and future validation requirements 35
3 OVERALL SOLUTION STRUCTURE 37
3.1 Integrated solution overview 37
3.2 High level architecture 37
1.3 Component 1 summary 38

1.4 Component 2 summary 38
1.5 Component 3 summary 39
1.6 Component 4 summary 39
1.7 End to end user and data flow 39
4 COMPONENT INTEGRATION AND COHESIVE OPERATION 41
4.1 Integration strategy 41
4.2 Interfaces data contracts and shared models 42
4.3 Shared nonfunctional requirements 44
4.4 Integrated workflow and exception handling 45
4.5 Team contribution and dependency matrix 46
5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION 48
5.1 Common technology stack and infrastructure 48
5.2 Development and integration process 49
5.3 Deployment and operational configuration 50
5.4 Evidence of team execution 51
6 INTEGRATED TESTING EVALUATION AND VALIDATION 53
6.1 Evaluation framework and success criteria 53
6.2 Integration and end to end testing 53
6.3 Overall results 53
6.4 Stakeholder and user validation 54
6.5 Achievement of common objectives 54
6.6 Integrated limitations and threats 54
7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION 55
7.1 Value proposition and beneficiaries 55
7.2 Market alternatives and differentiation 55

7.3 Business model and feasibility 55
7.4 Adoption intellectual property and compliance 56
7.5 Sustainable Development Goal contribution 56
8 CONCLUSION AND FUTURE DIRECTIONS 57
8.1 Overall conclusion 57
8.2 Principal integrated contribution 57
8.3 Future directions 58
REFERENCES 60
APPENDICES 62
Appendix A: Integrated interface specification 62
Appendix B Team contribution declaration 64
TEMPLATE CONVENTIONS ..... **Error! Bookmark not defined.**

LIST OF FIGURES

Figure 1 : High Level Architecture 38

LIST OF Table

Table 1 :Stakeholder Needs and Corresponding System Features..... 14
Table 2 : Sub Problems 25
Table 3: Group Objectives 28
Table 4 : Technical and operational constraints..... 33
Table 5 : Interfaces data contracts..... 42
Table 6 : Team contribution..... 46
Table 7 : Integrated Component Interface Specifications..... 62
Table 8 :Integration Validation and Error-Handling Requirements 63
Table 9 : Individual Research Team Contributions 64

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
ARIMA	AutoRegressive Integrated Moving Average
CC-CV	Constant Current–Constant Voltage
EV	Electric Vehicle
IEA	International Energy Agency
LLM	Large Language Model
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
RMSE	Root Mean Squared Error
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SOC	State of Charge
SOH	State of Health
SVM	Support Vector Machine
UN	United Nations
XGBoost	Extreme Gradient Boosting

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Electric vehicles (EVs) play an important role in the transportation sector's transition toward sustainability. The rapid development of batteries, artificial intelligence (AI), and machine learning (ML) enables building efficient, reliable, and user-friendly EV management systems. According to the Global EV Outlook 2026 report by the International Energy Agency (IEA), electric cars represented over a quarter of global passenger vehicle sales in 2025 [1]. This trend underscores the need for smart solutions to help EV owners manage battery operations, estimate the driving range, plan and control the charging process, and keep track of relevant technical information.

Despite the progress in EV management, drivers still experience several issues, including battery degradation, abnormal vehicle behavior, driving-range uncertainty, charging station availability, charging cost structure, and maintenance scheduling. Battery degradation impacts driving range, vehicle performance, reliability, and cost [2]. In addition, the ad-hoc approach to tracking individual battery parameters (e.g., state of charge [SOC]) might not be sufficient to capture overall battery health and identify abnormal operating conditions involving multiple parameters.

Conventional EV management approaches often involve separate applications or services for monitoring, battery-health assessment, charging-station discovery, route planning, and technical assistance. This approach can require users to consult several information sources before making a single vehicle-management decision. In addition, fixed-threshold warning mechanisms may not adequately represent the relationships among electrical, thermal, charging, motor, and driving parameters. Similarly, driving-range estimates can be affected by battery health, driving behavior, speed, temperature, elevation, and auxiliary power consumption. These aspects suggest the potential value of an integrated and intelligent approach to EV management.

Research has demonstrated the potential of machine learning to estimate battery degradation and state of health (SOH), identify abnormal operating patterns, predict

driving range, and forecast charging-station occupancy [3]– [10]. Personalized route-planning techniques can also take into account energy consumption, driving behavior, travel time, and charging requirements rather than simply geographical distance [11]. Furthermore, Retrieval-Augmented Generation (RAG) can combine retrieval with large language models to generate responses based on relevant knowledge sources, providing a basis for conversational EV assistance [12], [13]. However, these capabilities are often investigated as separate functions rather than as components of a unified vehicle-management solution.

To address this problem, this study introduces the concept of the IntelligentEV — an AI and ML-based mobile decision-support system comprising four interrelated components.

- The first component is the AI-Based EV Anomaly Detection System. This component analyses the battery voltage, current, and temperature, motor temperature, SOC, vehicle speed, acceleration, charging status, and power consumption data to detect operational inconsistencies. The system facilitates investigation and intervention by providing diagnostic insights with indicative abnormality levels and suggested courses of action.
- The second component is the Adaptive Battery Health Prediction and Personalized Recommendations. This component enables the user to stay informed about the battery health, with an emphasis on its state-of-health (SOH), including three-, six- and twelve-month projections. The system uses machine-learning algorithms to predict the battery's health trends and personalized rule-based advisory to suggest optimal charging practices, thus facilitating informed battery management decisions.
- The Third component, the Intelligent EV Charging Optimization and Range Prediction System, is designed to help the user to optimize the EV's utilization by predicting driving ranges, with a focus on the range around the vehicle, evaluating the charging station availability, wait time, route, charger selection, and cost. By analyzing the vehicle's characteristics, power consumption data, charging needs, and related travel requirements, this

component assists in making smart and informed decisions regarding the charging and utilization of the vehicle.

- The fourth element, AI-Based EV Virtual Assistant Using Retrieval-Augmented Generation (RAG) with Voice Support, is a conversational agent that allows users to access information related to their EVs virtually and verbally. It uses an EV knowledge base, information retrieval, RAG, and large language model to provide context-aware guidance on battery, charging, maintenance, and driving.

The four components described above represent a unified platform that enables diagnostic, predictive, optimization, and conversational functionalities. This solution aims to facilitate a less fragmented experience for managing EVs and enable users to make informed decisions about their EVs' usage and management. It is noteworthy that the recommendation feature is not intended to replace manufacturer-authorized diagnostic and maintenance services but to provide a helpful guidance tool.

1.2 Stakeholders and user needs

The proposed IntelligentEV system is intended to address the needs of various stakeholders, including the EV owners and drivers, as the primary users, and the EV service technicians, charging stations maintainers, and administrators, who are the secondary users. The main stakeholders are EV owners and drivers since they will use the information provided by the system. The system will provide information to EV owners and drivers about the vehicle's status, battery performance, driving range, and advice on making charging decisions. Furthermore, the system will serve secondary stakeholders, such as service technicians and mechanics, providing them with information about the vehicles' state, and charging station maintainers and administrators, who will use the information to monitor and control the charging stations' performance. The stakeholders were selected based on the system's requirements and proposed objectives. However, the stakeholder list might need to be reviewed and updated based on stakeholder feedback and testing results.

Primary stakeholders

Electric vehicle owners and drivers are the target audience for this proposed mobile application. This software will be helpful to them by providing a convenient way to get an overview of their car's condition, calculate the driving range, find charging stations, predict the time and cost of charging, and get recommendations on battery maintenance. In addition, users might be interested in the opportunity to obtain data on the technical aspects of electric vehicles without having to understand their working principles. The ability to interact with the app in both text and voice modes is another advantage of the proposed product.

Secondary stakeholders

- EV service technicians and maintenance providers: Might be interested in diagnostic data such as the detected categories and severity of anomalies, parameters that contribute to them, and alerts that could allow investigating possible issues.
- Charging-station operators: Could use the forecasts of station occupancy and queues, along with the availability of chargers, to optimize the charging process. This functionality would depend on the operators' access to the corresponding data.
- System administrators and developers: Would benefit from diagnostic data, model performance reports, and knowledge base updates to maintain the app's high quality.
- EV manufacturers and battery-management professionals: Could potentially use battery forecasts and their impact on operations to gain insights into the battery health and maintenance needs, depending on the availability of the required data and the possibility of its integration.

The main needs of the users and the features that the system should provide to satisfy these needs are presented in Table 1 below.

Table 1 :Stakeholder Needs and Corresponding System Features

Stakeholder	Identified need	Proposed system feature
EV owners	Understand vehicle status and define abnormal conditions	AI-based EV anomaly detection
EV owners	Keep track of the battery's condition and battery-care recommendations	Battery SOH prediction and personalized recommendations
EV owners	Predict the range and optimize trips	Intelligent range prediction and route optimization
EV owners	Find the charging stations and predict waiting time	Charging-station availability and queue prediction
EV owners	Predict the charging time and cost	Charging-time and cost estimation
EV owners	Get access to the EV-related information	RAG-based virtual assistant with voice/text support
Service technicians and providers	Interpret abnormal conditions	Diagnostic alerts, anomaly categories, and severity levels
Charging station operators	Get information on the anticipated traffic	Charging-station occupancy and queue prediction
System administrators and developers	Ensure reliable information and system performance	Data preprocessing, model evaluation, and knowledge base maintenance

The proposed system would meet these needs by providing a unified mobile decision-support platform instead of forcing the user into relying solely on separate apps for different tasks. The anomaly detection component would allow for identifying unusual circumstances, while the battery health component would provide long-term forecasts and behavior-based recommendations. The charging and range component would then assist in planning mobility, and the virtual assistant would be available to offer an accessible interface to the user for EV-related information.

While these features would be beneficial in terms of presenting the necessary information about the vehicle's status in a more accessible way, the quality of the decisions the user can make relying on this information depends on how good the data underpinning the system is, how accurate the models are at any given point, how much external information is available, and how relevant the produced suggestions are. Therefore, the system should be seen as a decision-support tool rather than a complete substitute for professional vehicle maintenance or manufacturer-provided diagnostics.

1.2 Alignment with the Sustainable Development Goals

IntelligentEV research project will contribute to several UN Sustainable Development Goals (SDGs). Thus, it addresses the need to create intelligent systems for managing electric vehicles, minimizes their energy consumption during charging, and uses AI and ML methods for transportation-related purposes. It can have an impact on these goals by introducing the corresponding technologies and methods, which will allow making more optimal decisions when managing EVs; however, this contribution will depend on how the system will work and be used in practice.

- **SDG 7: Affordable and Clean Energy**

SDG 7 goal and targets relate to the IntelligentEV project because optimizing energy use is an essential consideration in EV operations. The charging optimization and range-prediction component estimates driving ranges, predicts charging-station queues, recommends appropriate routes and charging stops, and estimates charging time and cost. These features contribute towards more energy-conscious EV

operation at a user level. The battery-health prediction component also facilitates more effective battery management through its SOH forecasts and battery-care personalization features. Overall, the features can be considered as facilitating energy-efficient EV operations and helping users make informed choices on charging and routes based on predicted energy needs. However, the project does not itself directly address improved renewable energy use, decreased electricity consumption, or affordability at a population level, and thus requires further assessment and measurement.

- **SDG 9: Industry, Innovation and Infrastructure**

SDG 9 directly benefits from the contributions of the project as it introduces innovative technological solutions to a variety of EV-related issues. IntelligentEV utilizes AI, ML, predictive modelling, optimization algorithms, and RAG-based conversational assistance to address multiple aspects of EV operation and management. By implementing the described methods in a single application, the project aims to provide a convenient and integrated experience offering services such as anomaly detection, battery-health prediction, charging and range optimization, and technical assistance.

The anomaly detection component analyzes the data provided by the vehicle to identify abnormal patterns resulting from the vehicle's improper operation. The battery-health component employs predictive modeling to estimate the SOH (State of Health) of the EV battery with a forecast interval of three, six, and twelve months ahead. The component of the application dealing with charging and range uses forecasting and route search methods to provide information and advice on the mobility of the EV. Finally, the virtual assistant utilizes retrieval-based methods to supply the user with the required information related to EVs. Therefore, the application encompasses various innovative technologies related to the fields of transportation and advisory systems.

The contribution of the project to Goal 9 is evident considering the variety of technological solutions employed. However, the actual impact on the sustainable

development goal has not been evaluated as the effectiveness of the proposed solutions has not been proven via prototype testing.

- **SDG 11: Sustainable Cities and Communities**

SDG 11 is indirectly addressed by the project's focus on accessible and informed transportation management. One of the challenges faced by EV drivers is the coordination of driving range, charging-station availability, expected waiting time, and route selection. These considerations become particularly important while planning journeys that require charging stops, as well as during the times of charging facilities' limited availability.

The proposed Intelligent EV Charging Optimization and Range Prediction System aims to address these challenges by way of range estimation, charging-station selection, queue forecasting, and route optimization, and by presenting the relevant information to the user in a single mobile interface.

The anomaly-detection component of the AI-based system also provides diagnostic information that allows users to identify unusual vehicle conditions and decide whether further investigation is needed. However, the project has not demonstrated its capacity to reduce traffic congestion, improve road safety, and increase access to sustainable transportation on a city-wide scale. These are areas where the project's impact would tangentially address the goals set for SDG 11, but they are not the project's direct impacts.

In general, the project's strongest direct alignment is with SDG 9, through technological innovation, followed by SDG 7, through energy-aware EV management, and SDG 11, through intelligent mobility planning. In all cases, the project's contributions are projected to come from the intended technical capabilities and the decision-support functions they enable, while the broader environmental, economic, and societal benefits require further validation

1.3 Purpose and scope of the common report

The purpose of this common research report is to describe the context of the general problem, motivation, integrated system design, and expected contribution of the

Electric Vehicle Management System Using Artificial Intelligence and Machine Learning. This document provides an overview of the proposed IntelligentEV platform and explains how each of the four individual research components fits into the bigger picture.

This report establishes the common research problem that emerges from the fragmentation of existing functions related to EV management. It describes the needs of EV users and related stakeholders, explains how the project can contribute to relevant Sustainable Development Goals, and provides an overview of the scope of the proposed system. Moreover, it gives a common context for the individual components, including design considerations, technologies, evaluation approaches, and limitations.

The actual research and development activities are organized into four components that address individual aspects of EV management.

1. AI-Based EV Anomaly Detection and Classification: This project focuses on developing an algorithm for analyzing vehicle operating parameters, identifying anomalies, classifying them according to severity, and providing information for diagnostics and further analysis.
2. Battery Health Prediction and Personalized Recommendation: This project focuses on creating an algorithm for predicting battery state-of-health (SOH) in three-, six-, and twelve-month forecasts and personalized recommendation systems based on charging and driving patterns.
3. Intelligent EV Charging and Range Prediction: This project focuses on the creation of an algorithm for driving-range estimation, prediction of charging station availability and waiting time, optimized route planning with charging stations, and estimating the time and cost of charging.
4. AI-Based EV Virtual Assistant: This project focuses on the creation of a knowledge-grounded EV virtual assistant based on retrieval-augmented generation (RAG) with voice support.

Each single thesis presents the results of the detailed investigation of the specified problem. All the four theses contain the analysis of the literature sources, data,

algorithms, systems, test results, and possible limitations of the solutions. Each single study presents the technical information that proves the possibility of developing the given solution.

Meanwhile, the common report focuses on the relation between the four components. The four components of the mobile decision-support system for EV owners and drivers are operational diagnostics, battery-health forecasting, charging and range optimization, conversational guidance. All the four components complement each other to achieve best results in the single system.

The scope of the common report of the whole project comprises the following scopes:

- The background and context of developing intelligent EV management and the common problems EV user's encounter.
- The stakeholder's needs that will be addressed and their association with relevant Sustainable Development Goals.
- How to combine all four AI and ML components into a single mobile application.
- The technologies, data, and methods that will be used for developing the technology.
- The analysis and assessment of the system's possible evaluation, limitations, and directions of development.

The project can be limited in several aspects. Firstly, the results depend on the quality of the data sets, test scenarios, prediction accuracy, and availability of the required information on cars and charging stations. Secondly, the recommendation engine relies on the set of predefined behavioral rules, and the virtual assistant's results depend on the coverage and freshness of the knowledge base related to EVs. Finally, there is a need for further testing on larger scales and more diverse sets of users, vehicles, and scenarios to confirm the practical implementation and benefits of the system.

Therefore, the common report introduces the IntelligentEV project as an integrated initiative that unifies all four solutions and implies their combined research and development efforts. It defines the overall approach, describes the parts' common basis, and highlights the importance of separate studies to provide a consistent perspective on each aspect of the research.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

Improper management of electric vehicles leads to negative consequences, such as a lack of information on the vehicle's operating conditions, the state of batteries, operating range, charging needs, and specific maintenance issues. These aspects of management differ from each other in terms of information, analysis, and decision-making. Separating the information about the occurrence of anomalies, the state of health of batteries, charging needs, and technical information makes it difficult for users to organize, understand, predict, and perform actions.

The main problem that the Electric Vehicle Management System Using Artificial Intelligence and Machine Learning project solves is a complicated way of providing decision-making support information on the interrelated operational, battery-health, charging, and information needs of electric vehicles.

This problem has several aspects that are of practical interest. First, identifying which vehicle operating conditions are abnormal requires distinguishing unusual situations from normal variations. Second, battery aging plus variable energy consumption patterns makes it difficult to estimate future battery condition and the driving range of an electric vehicle. Third, charging decisions depend on the relation between the vehicle's remaining energy, the destination, and the availability of charging stations, including waiting time and cost. Finally, users need intelligible system explanations and relevant information to interpret the messages and respond appropriately.

The problem aspects are not independent and should be considered together. For instance, abnormal vehicle operating conditions may affect energy consumption and, therefore, the driving range; battery-health deterioration contributes to reduced capacity; and waiting time at a charging station may dictate the need for a detour. In addition, the value of in-vehicle predictions and alerts depends on the driver's ability to comprehend the information and act upon it.

Thus, the common research problem is not only the lack of individual EV monitoring or prediction function. It is the challenge that multiple connected EV management requirements form a coherent decision-support framework. The proposed IntelligentEV platform explores this challenge in four complementary research components, including operational anomaly detection (i.e., operational EV monitoring), charging optimization and range prediction (i.e., operational EV planning), battery-health prediction and personalized recommendation (i.e., operational EV recommendation), and a RAG-based virtual assistant with voice support.

2.2 Root causes and overall research gap

The main issue is caused by several interrelated technical and practical limitations in EV management. While there are individual technologies for vehicle diagnostics, battery control, charging management, route planning, and user assistance, the capabilities suggested in the research paper address different aspects of the EV management process. The overall issue is to integrate these capabilities in a unified framework while maintaining the specific analytical needs of each process component.

2.2.1 Root causes of the problem

1. Dispersed Vehicle Information and Monitoring Capabilities

Due to the numerous operating parameters and information resources related to electric vehicles (EV), the current monitoring system cannot provide a holistic view of the EV status. Monitoring the battery status cannot give a comprehensive idea of the motor status or unusual power draw. Also, it is unclear if the diagnostics and data about the charge and discharge are available in the same system or if they have to be retrieved in separate tools.

2. Limited ability to identify unfamiliar operational abnormalities

Conventional diagnostic approaches often rely on predefined fault codes or threshold-based warnings. While these are effective for known conditions, they can prove inadequate in identifying unfamiliar or subtle deviations in complex operational data sets. The research paper highlights the importance of investigating

the application of anomaly detection techniques which operate by learning normal operating patterns and then flagging unusual observations for further investigation.

3. Uncertainty in battery degradation and future vehicle performance

Battery state of health changes over time while range is highly dependent on state of charge, speed, temperature, elevation and auxiliary power consumption amongst other factors. A single battery or range figure cannot be assumed to be sufficient in determining future vehicle performance. Predictive models require appropriate historical data, relevant input variables and consideration of relevant prediction horizon

4. Complexity of charging and travel decisions

The process of picking a charging station or route is not as simple as picking the closest available charging station. Some consideration of personal range requirements, expected energy consumption, station occupancy, waiting time, power rating, tariffs and personal preferences are required before making the decision to pull over and charge. By not considering all these constraints the "best" route may not be selected.

5. Limited personalization and technical support

Generic EV advice may not account for personal driving and charging patterns, driving distance and overall driving behavior. At the same time technical information may not be intuitive and may require additional explanation for the average EV owner. Personalized advice and EV-specific conversational interface could help bridge the information gap, provided the sources of information are reliable.

2.2.2 Overall research gap

The overall research gap is the need for a coordinated EV management approach, which addresses four inter-related requirements: detecting abnormal vehicle operation, predicting battery and driving performance, supporting charging and route decisions, and presenting relevant information in an accessible form.

The research paper motivates these functions by separate component level methods: Anomaly detection explores Isolation Forest, One-Class Support Vector Machine (One-Class SVM) and Deep Autoencoder approaches. Battery-health prediction compares machine-learning models and includes a recommendation mechanism. Charging and range management combines predictive methods with route optimization and charging calculations. The virtual assistant uses RAG to retrieve EV-related information and generate conversational responses.

The gap addressed by the common project is therefore, the integration of complementary analytical and user-support capabilities into a single EV management framework, rather than the invention of each underlying algorithm independently.

The project also faces an evaluation challenge: a strong result in one component does not demonstrate that the complete system provides reliable end-to-end decision support. For example, accurate range prediction does not show that the selected route is optimal under real-world charging conditions. Similarly, anomaly detection does not establish that a recommended maintenance action is correct. The four components therefore need their own appropriate evaluation methods, and an assessment of how their outputs can be used together.

An intended contribution is to consider whether an integrated platform can make operational information more useful for EV users and related stakeholders. Claims about improved safety, reduced costs, longer battery life, or environmental benefits must remain objectives for future validation, unless they are supported by direct measurements.

2.3 Subproblem and component mapping

The common problem can be decomposed into four subproblems. Each of them has a specific research aim, expected result, and evaluation requirement. These four subproblems are complementary to each other since they target at different deficiencies, and the combination of their results can achieve the overall goal of managing the intelligent EVs.

Table 2 : Sub Problems

Subproblem	**Specific problem**	**Corresponding component**	**Expected output**
SP1: Operational anomaly identification	Unusual vehicle behavior's may not be accurately identified by diagnostic rules due to the need to consider various factors during diagnostics.	AI-Based EV Anomaly Detection system	Anomaly scores, categories, severity, parameters affected, alerts and suggested actions.
SP2: Charging and journey uncertainty	Driving range, charging delays, suitable routes and charging time or cost may not be accurately estimated by drivers.	Intelligent EV Charging Optimization and Range Prediction System	Predicted range, estimated queue time, suggested route, and estimated charging time and cost.
SP3: Battery-health uncertainty and generic guidance	Battery degradation and future battery condition may not be understood by	Adaptive Battery Health Prediction and Personalized Recommendations	Estimated battery SOH, predictions of future battery health and personalized

	users and guidance may be generic and not consider users' usage.		recommendations.
SP4: Difficulty accessing and interpreting EV information	EV information may not be found easily and may not be interpreted by users easily, especially when a contextually appropriate explanation is needed.	AI-Based EV Virtual Assistant Using RAG with Voice Support	Answers retrieved or generated based on context provided will be presented in text or voice.

Interdependence between the four subproblems

The components answer slightly different questions related to the same EV management process.

- SP1 — What is abnormal? The anomaly detection component analyzes the vehicle operating data and provides information that may need investigation.
- SP2 — What should we consider for the journey or charging decision? The charging and range component evaluates travel and charging conditions to support route and charging decisions.
- SP3 — How could the battery status change, and what practices could it improve? The battery-health component estimates SOH and uses the available battery and use information to provide recommendations.

- SP4 — How can the users understand the information and get relevant guidance? The virtual assistant offers a conversational entry to the EV knowledge base.

These links establish a logical connection between the subproblems, but they do not replace each other. An anomaly score is not a battery-health estimation, a battery-health estimation is not a driving-range prediction, and a conversational answer is not a substitute for the underlying analytical results or a confirmed vehicle diagnosis.

The components may support each other at the level of the information and user decisions, but their actual technical integration will depend on the system's architecture and available data interfaces. Any claim that the outputs are automatically shared between components must be supported by the implemented design and testing.

2.4 Overall aim and group objectives

2.4.1 Overall aim

The overall objective of the project was to design, develop, and evaluate an integrated electric vehicle management system that would apply artificial intelligence, machine learning, optimization processes, and retrieval-augmented generation to facilitate operational anomaly detection, battery-health prediction, charging and range management, and accessible EV related decision support.

The project aimed to unify the four aspects of the research under one objective but retain the technical responsibility of each component. The focus was on the development and evaluation of a system that could convert available vehicle data and EV-related knowledge into a consumable format that could help users make informed decisions.

2.4.2 Group objectives

The following objectives have been defined to structure the overarching objective into verifiable activities involving the development and component-level evaluation of the proposed solution.

Table 3: Group Objectives

Objective ID	Group objective	Component mapping	Evidence used for evaluation
GO1	Develop and evaluate an AI-based approach to detect abnormal EV operating conditions by analyzing the relevant vehicle parameters and comparing the results of applicable anomaly detection models	SP1 — Anomaly detection	Model comparison, detection of normal-abnormal scenarios and severity levels, and the correctness of diagnostic results when ground truth data is available
GO2	Develop and evaluate methods for range prediction, waiting time estimation for the charging station, route optimization, and calculation of charging time and cost	SP2 — Charging and range prediction	Range prediction error, queue prediction error, route planning evaluation, and the correctness of calculated charging time, and cost

GO3	Develop and evaluate the battery SOH prediction method and a recommendation module based on battery and user-specific data	SP3 — Battery health and personalized recommendations	Prediction accuracy across the chosen prediction horizons, comparison of the candidate models, and assessment of recommendation relevance and consistency with the available data
GO4	Develop and evaluate the EV-specific virtual assistant capable of retrieving the relevant information and simulating the question-answering dialogue	SP4 — RAG virtual assistant	Information relevance, factual correctness of the generated responses, coverage of EV-related queries, response usefulness, and voice recognition accuracy
GO5	Establish the unified system framework that would integrate all components and provide the end-user with the required decision-	All components	Functional tests, component-specific tests, end-to-end tests if available, and the assessment of the application usability

	support functions in a consistent application environment		
GO6	Define the technical limitations, data requirements, and validation constraints of the proposed solution that are critical for the practical implementation and application of the developed solution	All components	Analysis of data coverage, model-specific limitations, and validation results across the different test sets and identification of the primary technical and data-related challenges.

2.4.3 Evaluation strategy and success criteria

Each objective requires a different type of evidence suitable for the problem to be solved, and a single evaluation metric would not be adequate for all four components since they yield different types of output.

For anomaly detection, the evaluation should concern the ability to discriminate normal operational behavior from anomalies and, where available labels, the accuracy of anomalous classifications. For charging and range prediction, quantitative prediction errors and the performance of the route and charging calculations present relevant evidence. Battery-health prediction needs to compare predicted and observed SOH values, while personalization should be evaluated separately from quantitative prediction accuracy. The virtual assistant needs

evaluation of retrieval relevance, answer accuracy, knowledge coverage, and voice interaction.

The research paper reports component-level prototype results, including range-prediction errors, queue-prediction error, battery-health model metrics, and evaluations of route and charging calculations. These results can inform the evaluation of the relevant objectives, but they should not be interpreted as proof of complete real-world system performance.

The common project should therefore distinguish between component-level success and overall system success. Component-level success concerns whether each function performs adequately under its test conditions. Overall system success additionally concerns that the components work together as intended, provide understandable outputs, and support realistic user scenarios. Quantitative thresholds for success should be defined according to the available datasets, experimental design, and requirements of each component rather than selected arbitrarily.

The scope of the common project defines the functions that the proposed IntelligentEV system is intended to tackle, the assumptions under which its components will operate, and limitations which must be taken into account when interpreting its results.

2.5 Scope assumptions and constraints

2.5.1 Scope of the complete solution

The project encompasses the design, implementation and evaluation of four related EV management capabilities:

- Analysis of vehicle operating parameters to detect abnormal conditions and generate diagnostic information.
- Prediction of driving range and charging-station waiting times, plus route optimization and charging time and cost estimation.
- Estimation of battery SOH over selected prediction horizons and recommendations based on available usage information.
- Delivery of EV-related information in the form of a RAG-based virtual assistant with text and voice support.

The common scope also includes the high-level connection between these components, their intended users, and the evidence needed to evaluate their performance. Detailed model development, dataset preprocessing, implementation decisions, and component-specific experiments belong mainly in the corresponding individual research reports.

2.5.2 Scope assumptions

The following assumptions define the conditions under which the intended functions of the system can be developed and evaluated.

- Availability of relevant data: The analytical components require suitable historical or operational data. The accuracy and usefulness of their outputs will depend on the relevance, completeness and quality of these inputs.
- Representativeness of datasets: Datasets used for model development are assumed to contain information relevant to the particular prediction or detection task. Performance on a particular dataset does not necessarily indicate generalizability to other vehicles or operating environments.
- Availability of charging information: Queue prediction and charging-related decisions will rely on access to suitable information about the status of chargers, historical charging sessions, availability data, and relevant charging parameters.
- Availability of user-usage information: Personalized battery recommendations will rely on the usage variables available to the recommendation mechanism. Recommendations cannot reliably reflect personal behavior that has not been observed or provided.
- Availability of relevant knowledge: The RAG-based assistant will rely on an EV-specific knowledge base that contains information appropriate to the questions of its users. Answers provided by the assistant will be incomplete or out-of-date if the underlying knowledge sources are insufficient.
- Human interpretation of system outputs: Alerts, predictions and recommendations will be intended to assist users. Users and relevant professionals will remain responsible for appropriate decisions, especially where safety or maintenance is concerned.

These are operating assumptions for the proposed system rather than claims than all required data sources or integrations are continuously available in a deployed environment.

2.5.3 Technical and operational constraints

Table 4 : Technical and operational constraints

Constraint	Description	Implication for the project
Dataset limitations	The available datasets may not include all possible vehicle models, battery chemistries, driving conditions, climates, and failure scenarios.	Results should be considered applicable for the conditions represented in the evaluation data
Anomaly ground truth	Determining whether an anomalous observation represents an actual fault may require reliable labels or expert assessment.	An anomaly score should not be presented as a confirmation of a diagnosis.
Battery-health prediction	The prediction of future SOH relies on the quality of historical battery behavior and data, and the prediction horizon.	Longer-term predictions require appropriate validation and consider communicating uncertainty where possible.
Range and queue variability	Driving conditions, energy consumption, charger availability, and waiting time can affect driving performance and impact	Predictions may become less reliable where operating conditions differ from the available data.

	variations in the predictions.	
Route and charging dependencies	The suggested route relies on the accuracy of the underlying network, charging information, energy estimates, and cost inputs.	Incomplete or out-of-date information can impact that value of the suggested route.
Recommendation-rule limitations	Personalized recommendations may rely partly on predefined behavioral rules rather than a fully learned personalization model.	Recommendation quality should be considered independently of battery-health prediction performance.
RAG knowledge coverage	Generated responses are limited by the relevance and timeliness of retrieved material and the language model's behavior.	Incorrect, incomplete, or unsupported answers are possible and require appropriate safeguards.
Voice interaction	Recognition performance can vary with background noise, pronunciation, speech clarity, and language differences.	Voice support requires independent testing with representative user conditions.
Integration and usability	The integration of several analytical components has additional interface, data-format, response-time, and consistency requirements.	Each component should be tested individually and, where implemented, in integrated user workflows.

2.5.4 Exclusions and boundaries

In order to create a viable, research-focused system, the common project does not assume that the suggested system handles all functions related to a full commercial EV management platform.

Unless directly shown in the implemented component examples, the project should not be advertised as being able to offer:

- Direct control of battery charging hardware, vehicles' power systems, or motor controllers.
- Certified fault diagnosis or guaranteed detection of all potential vehicle failures.
- Guaranteed driving range, battery lifespan, charging availability, or future battery condition.
- Universal compatibility with all EV manufacturers, battery technologies, and charging networks.
- Guaranteed real-time access to every charging station and external-service connectivity.
- Autonomous emergency response or replacement for professional maintenance and manufacturer safety instructions.
- Independently-verified reductions in emissions, ownership costs, charging delays, and battery degradation.

These factors distinguish an intelligent decision-support research proof of concept from a fully verified commercial vehicle-control or maintenance system.

2.5.5 Dependencies and future validation requirements

The practical applicability of the full solution depends on many factors apart from implementing individual machine-learning models: availability of representative data sets, reliable operational and charging information, appropriate application

interfaces, communication between components, and a maintained EV knowledge base.

Further validation should explore more vehicle models and battery chemistries, a wider set of driving and environmental conditions, previously unseen anomalies, and shifts in charging-station demand. The virtual assistant should be tested for factual consistency, retrieval failures, and voice-recognition limitations as well. Finally, integrated testing should explore realistic cases where the user has to interpret a vehicle warning, account for its possible effects on the trip, find alternatives for charging, and receive appropriate recommendations.

The scope and constraints delineated in this section provide a common basis of assessment for the four components without overstating their capabilities. Furthermore, it establishes an important distinction between the evidence of the project's current experimental viability and what additional testing is needed before the system could be relied upon for broader real-world applications.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

The proposed Electric Vehicle Management System using Artificial Intelligence and Machine Learning, known as the IntelligentEV, is an integrated decision-support platform that harmonizes four EV management functions into a common system. Instead of treating vehicle anomaly detection, battery-health prediction, charging optimization, and user assistance as separate services, the platform aims at making their outputs accessible to users within a common environment.

The system uses AI and ML models, optimization techniques, battery-health analysis, and a retrieval-augmented generation (RAG) virtual assistant to help users understand vehicle conditions, anticipate potential issues, plan journeys and get EV-related information. Its key value proposition lies in transforming available vehicle data and EV-related knowledge into comprehensible information. The platform supports human decision-making but does not replace professional diagnosis or manufacturer-approved maintenance procedures.

3.2 High level architecture

The high-level description of the architecture is represented by the user-facing application, shared data and application services, four intelligent components, and relevant data sources or external services. End-users utilize the application to access the results of analytics, recommendations, charging information, and conversation. The components are using specific inputs and applying corresponding methods to process them. The outputs are further provided via the application, and external data and knowledge sources supply information when possible. The level of immediate connection and communication is defined by the actual interfaces and services implemented in the chosen solution.

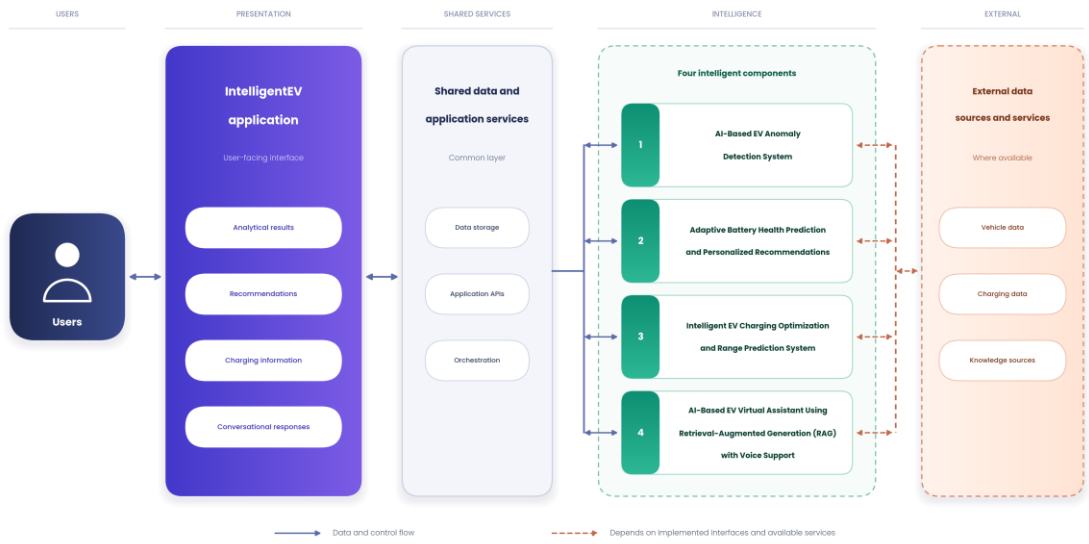


Figure 1 : High Level Architecture

1.3 Component 1 summary

- Research question: Identification of abnormal vehicle operating conditions for which the user could benefit from specific diagnostic rules.
- Inputs: battery voltage, current, temperature, motor temperature, state of charge (SOC), speed, acceleration, charging status, and power.
- Core methods: Isolation Forest, One-Class SVM, and Deep Autoencoder.
- Outputs: Anomaly score, category, severity, parameters, alerts, and recommended actions.
- Contribution: Provide awareness of abnormal vehicle behaviors that may require additional diagnostics attention.

1.4 Component 2 summary

- Research question: Reduction of range anxiety, charging waiting time, route optimization, and charging cost.
- Inputs: SOC, speed, ambient and battery temperature, height, auxiliary power consumption, charger status, session history, and tariff.
- Core methods: Hybrid range prediction model (uses Long Short-Term Memory networks combined with regression), queue prediction algorithm (ARIMA-LSTM hybrid model), and route optimization algorithm (Graph-based and uses A or Dijkstra).

- Outputs: Predicted range, waiting time, route recommendations, and charging time/cost suggestions.
- Contribution: Assist users in understanding the current situation and predicting future states of their vehicles.

1.5 Component 3 summary

- Research question: Estimation of battery degradation with personalized recommendation based on user behavior patterns.
- Inputs: The battery's capacity history and users' availability records, including the number of charging sessions, fast-charging sessions, daily mileage, and driving patterns.
- Core methods: Battery state-of-health (SOH) estimation, comparison of Random Forest and XGBoost, and rule-based recommendation algorithms.
- Outputs: Estimated battery SOH, recommendation for the selected period, and corresponding battery-care advice.
- Contribution: Enable users to evaluate the potential battery degradation and understand which behavior patterns are battery friendly.

1.6 Component 4 summary

- Research question: How can EV-related information be better organized to facilitate communication with end-users about their EVs?
- Inputs: Natural language questions asked by the user and corresponding answers retrieved from the knowledge base.
- Core methods: Retrieval-augmented generation, language model fine-tuning, and speech-to-text/text-to-speech algorithms (if applicable).
- Outputs: User-friendly answers to the EV-related queries in natural language.
- Contribution: Create an accessible information platform that allows end-users to quickly navigate and understand EV-related information.

1.7 End to end user and data flow

The general idea is that when a user interacts with IntelligentEV or the related data about vehicles, charging, or past events becomes available in the shared application

services. These services then route the information to the specific component that handles the requested task.

- Input: The application receives vehicle operating data, battery information, charging details, or user questions from a particular source.
- Intelligent analysis: The selected component analyzes the input and identifies any irregularities, determines how far one can drive with the current battery power, predicts charging requirements, or provides knowledge about EVs.
- Output: The system then prepares alerts, predictions, suggested routes, recommendations, or answers for the user.
- Presentation: The application displays the results in a way that is easy to understand and allows the user to view and respond appropriately.
- User response: The user applies the information to vehicle operation, route planning, charging selection, battery maintenance, or additional research.

The four components support the process at different stages. The anomaly detection assists in operating the vehicle, while the other three elements help with driving decisions. In particular, the route and charging recommendation component predetermines travel options, the battery health predictor analyzes long-term trends, and the virtual assistant provides a simple interface to obtain information. Thus, all parts of the application are interconnected and operate within a shared decision-making paradigm but perform specific tasks.

It is emphasized that the accuracy of the results depends on the quality of the input data, the correctness of the models, the availability of third-party services, and the completeness of the knowledge base. For this reason, the results should be considered informative and applied with caution.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

IntelligentEV is designed as one mobile decision-support system that brings together four related capabilities: (1) AI-Based EV Operational Anomaly Detection, (2) Adaptive Battery Health Prediction and Personalized Recommendations, (3) Intelligent EV Charging Optimization and Range Prediction, and (4) AI-Based EV Virtual Assistant Using Retrieval-Augmented Generation (RAG) with Voice Support. The objective of integration is to give an EV user a consistent place to understand vehicle condition, anticipate battery-health changes, plan journeys and charging, and obtain contextual EV guidance. The system addresses the need to make these functions available through one coordinated solution [1].

The components have distinct responsibilities but share a common EV context. Component 1 analyses operational measurements such as battery voltage, current, temperature, state of charge (SOC), speed, acceleration, charging state and power consumption. Component 2 uses historical battery observations to forecast State of Health (SOH) over 3-, 6- and 12-month horizons and combines these forecasts with behavioural rules to produce battery-care recommendations. Component 3 uses vehicle, route and charging-station information to estimate remaining range, predict charging queues, select charging stations and routes, and estimate charging time and cost. Component 4 retrieves relevant information from an EV-specific knowledge base and uses an LLM to formulate text or voice responses about battery care, charging, maintenance and efficient driving.

A suitable integration sequence is to agree on the shared user interface and data definitions first, connect each component through a documented interface, and then verify each component independently before performing end-to-end tests. Integration should proceed in increments: (i) confirm each component can accept valid inputs and return a documented output; (ii) connect the component outputs to the common application layer; (iii) test that user and vehicle context is passed consistently; and (iv) test normal, missing-data, unavailable-service and invalid-input scenarios. This

interface-first approach reduces the risk that components use incompatible field names, units, identifiers, or assumptions.

The system brings the four components together within one mobile application and evaluates their component-level performance. The architecture described in this report does not assume a particular API gateway, microservice framework or continuous-integration pipeline unless that choice is confirmed by the implemented system and its configuration.

4.2 Interfaces data contracts and shared models

The four components need consistent data contracts so that their outputs can be presented together without confusing the user. At the conceptual level, the shared vehicle context should identify the vehicle or test record, the time of observation, the source of the data, and the relevant units. Each component should validate its input before processing and return a predictable response containing the result, status, timestamp and any warnings. Where a field is unavailable, the interface should represent it explicitly as missing rather than silently treating it as zero.

The following interface contracts are recommended for the integrated solution. They define logical exchanges based on the inputs and outputs required by the four components; they should not be interpreted as existing endpoint names unless those names are confirmed in the implementation [1].

Table 5 : Interfaces data contracts

Interface / component	Key input data	Expected output	Validation and error handling
Operational anomaly detection	Timestamp; voltage; current; battery and motor temperatures; SOC; speed; acceleration; charging state; power consumption.	Vehicle status; anomaly score; anomaly category; severity; affected parameters; alert and recommended action.	Check units, required fields, plausible ranges and timestamp validity. Return a validation error for malformed records; do not label missing measurements as normal.
Battery SOH prediction and recommendations	Historical battery observations and capacity-related	SOH estimate or forecast for 3, 6 and 12 months; model/version	Check feature schema and units; identify unsupported horizons and insufficient

	features; prediction horizon; user charging/driving behaviour where available.	metadata; personalized battery-care suggestions.	history. Clearly indicate when a forecast cannot be produced.
Charging optimization and range prediction	SOC; driving speed history; ambient and battery temperature; elevation; auxiliary power; route/traffic context; charger status, power, tariffs and historical occupancy.	Estimated driving range; station queue/wait; recommended route and charger; charging duration and estimated cost.	Check SOC and charger data ranges, freshness of station/traffic data and availability of route inputs. If live data is unavailable, show that estimates are based on stored or supplied data.
EV virtual assistant (RAG and voice)	User text or speech transcript; relevant vehicle context if available; retrieved EV knowledge passages.	Context-grounded answer in text and, where supported, speech; relevant caveat or request for clarification.	Handle empty or unclear input, speech-recognition failure, retrieval failure and unavailable language-model service. Avoid presenting unsupported diagnostic statements as confirmed facts.
Shared application layer	User request; shared vehicle/test identifier; component request status.	Consistent dashboard, result summaries, timestamps and user-facing error messages.	Validate requests, apply access controls, handle timeouts and prevent one component failure from crashing unrelated functions.

The shared model should preserve units and semantics. For example, SOC and SOH are different measures: SOC represents the current charge level, whereas SOH represents battery condition relative to its rated capacity. Range should be expressed in distance units, queue estimates in time units, and charging cost in the currency associated with the tariff source. The application should include a timestamp or freshness indicator for data that may change, especially charger availability, traffic, and queue estimates. The integration layer should also record which model, or component version produced a result so that unexpected output can be investigated.

The actual transport mechanism—such as direct function calls, REST APIs or another interface—must be documented from the implementation. For every real interface, the final report should identify the endpoint or function name, request and response schema, authentication requirements, timeout, error codes, and versioning

approach. These implementation details must be verified from the project repository and configuration.

4.3 Shared nonfunctional requirements

Performance: The application should return dashboard information promptly and avoid making the user wait for every component when only one function is required. Model inference, route calculations, and retrieval-based responses should have defined timeout behavior. The team should measure response times using the actual implementation and report measured values rather than estimated targets as results.

Usability and accessibility: The four components should use consistent navigation, terminology, units, severity labels, and feedback. Results should be understandable to non-specialist EV users. Critical warnings should be visually distinct, while explanations should state what was detected, why it matters, and what action is recommended. Voice interaction should complement, not replace, the text interface.

Security and privacy: Vehicle telemetry, user behavior, location, voice recordings, and user queries may be sensitive. The implementation should collect only the data required for the requested function, restrict access to authorized users, protect data in transit and at rest where applicable, and avoid exposing personal or vehicle information in logs. Secrets and API keys should not be stored in source code. If voice audio is processed by an external service, users should be informed that data flow and the relevant retention settings should be checked.

Reliability and safety: Predictions are decision-support estimates, not substitutes for manufacturer warnings, qualified vehicle inspection or official charging-station status. The application should display uncertainty or data-freshness limitations where relevant. If a model or external data source fails, the system should report the limitation and avoid fabricating a result. A failure in the virtual assistant should not prevent access to the anomaly, battery-health or charging modules.

Maintainability and scalability: Component responsibilities should remain separate, with documented interfaces, configuration separated from code, and dependencies recorded. Model artefacts and datasets should have clear versions. Logs should

support debugging without unnecessarily recording personal data. The design should allow one model or external service to be updated without requiring unrelated components to be rewritten.

4.4 Integrated workflow and exception handling

During normal operation, the user opens the mobile application and selects a vehicle-related function. When operational measurements are available, the anomaly-detection component analyses the combined parameters and returns the status, anomaly score, category, and severity. Battery-history data can be passed to the SOH prediction component to obtain 3-, 6- and 12-month forecasts, while the recommendation engine uses available charging and driving behaviour to provide battery-care suggestions. For a journey, the charging-optimization component combines the available SOC, driving, route and charger information to estimate range, compare charging options, forecast waiting time, and estimate charging duration and cost. If the user needs an explanation, the virtual assistant retrieves relevant EV knowledge and produces a contextual answer through text or voice. The application presents the results in a consistent interface and makes clear which information is measured, predicted, or retrieved.

The integrated workflow must also handle exceptions. If required sensor or battery-history fields are missing, the relevant component should identify the missing inputs and request additional data or return a clearly labelled unavailable result. If charging station or traffic data cannot be retrieved, the route planner should indicate that current availability has not been verified and avoid presenting stale data as live. If a model fails or returns an invalid value, the application should log a diagnostic event and show a safe, understandable error message. If speech recognition fails, the user should be able to retry or switch to text. If knowledge retrieval returns no relevant material, the assistant should state that it cannot provide a well-supported answer instead of inventing EV-specific facts. Network interruptions should not erase already displayed results, and retry operations should be bounded to avoid repeated requests.

For safety-sensitive cases, such as a critical anomaly classification, the application should communicate the severity and recommend appropriate inspection or manufacturer guidance. It should not claim to remotely confirm a physical fault unless the available evidence supports that conclusion. The system includes anomaly scores and severity categories; the exact user-facing safety wording and operational escalation rules should be checked against the implementation [1].

4.5 Team contribution and dependency matrix

The project is divided among four individual components, with each student responsible for the component identified in the team allocation. Integration remains a shared responsibility because the value of IntelligentEV depends on the components working together through consistent interfaces and a unified user experience. The following matrix records the supplied allocation and the principal integration dependencies. Tasks beyond the named component should be confirmed by the team against repository history, meeting notes, and individual research diaries.

Table 6 : Team contribution

Team member / student ID	**Assigned component**	**Main responsibility**	**Integration dependency / evidence to confirm**
Vidushan A A D D N — IT22157928	AI-Based EV Anomaly Detection System	Operational parameter preparation, anomaly-detection approach, diagnostic output, and component-level evaluation.	Provides anomaly status, score, category, severity, and affected parameters. Confirm implementation details and test evidence from source code.
Fernando C T H H D — IT22247490	Adaptive Battery Health Prediction and Personalized Recommendations	Battery SOH forecasting across 3-, 6- and 12-month horizons and generation of behaviour-based battery-care recommendations.	Provides SOH forecasts and recommendations for the shared dashboard and assistant context where applicable. Confirm model artefacts and evaluation records.
Dias N. T. B. P. — IT22134080	Intelligent EV Charging Optimization and Range Prediction System	Driving-range estimation, charging queue/availability prediction, route and charger selection, and	Provides range, queue, route, charger and charging estimates. Confirm external data dependencies, units, and the actual code/test results.

		charging-time/cost estimation.	
Jayarathne W.T.D.K. — IT22071170	AI-Based EV Virtual Assistant Using Retrieval-Augmented Generation (RAG) with Voice Support	EV knowledge retrieval, LLM-based response generation, and text/voice interaction.	Consumes relevant user or vehicle context when available and returns contextual guidance. Confirm knowledge sources, retrieval pipeline, speech services and test evidence.

Shared responsibilities include agreeing on the application-level terminology and data contracts, reviewing cross-component compatibility, performing integration and regression tests, maintaining project documentation, and preparing the common report. The table identifies component ownership but does not, by itself, prove that a specific integration or testing task was completed by a particular student. The final submission should be supported by agreed contribution records and version-control evidence.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The proposed IntelligentEV solution follows a modular mobile decision-support architecture. A shared application interface presents the outputs of the four components, while the component logic performs anomaly detection, battery-health forecasting and recommendation generation, charging and range optimization, and retrieval-based conversational assistance. The exact mobile framework, programming languages, database engine, hosting provider, API framework, and version-control workflow must be documented from the actual project repository and configuration.

At the functional level, the infrastructure includes the following logical elements: (i) a mobile user interface for entering requests and viewing results; (ii) a component layer for the four analytical and AI functions; (iii) data storage or data access for historical battery observations, operational records and EV-specific knowledge; (iv) external or supplied mobility information for route, traffic, charger availability and tariffs where available; and (v) a model and configuration layer for storing the selected trained models and their required preprocessing settings. The exact physical arrangement—one application process, separate services or a hybrid deployment—should be described according to the actual implementation.

The system methods include Isolation Forest, One-Class SVM and Deep Autoencoder for anomaly detection; Random Forest and XGBoost regression for battery SOH forecasting, with Random Forest selected in the reported comparison; a hybrid LSTM and regression approach for driving-range prediction; ARIMA-LSTM for charging-queue prediction; A* and Dijkstra-based path finding for route selection; a CC-CV charging model for charging-time estimation; and RAG with an LLM for conversational assistance [1]. These are algorithmic choices, not a complete software-stack specification. The final report should separately list the actual libraries and versions used to implement each method.

For reproducibility, the project should record the source-code repository location, branch or release used for the demonstration, model and dataset versions, required environment variables, dependency versions, and setup instructions. Credentials should be configured securely and excluded from the repository. If an external API or hosted language model is used, the report should identify the service and its role without publishing secret keys.

5.2 Development and integration process

A suitable development process for the four-component system is incremental and interface-driven. Each component owner develops and evaluates the assigned component, documents its inputs and outputs, and supplies a reproducible way to run the component. Before connecting components, the team agrees on common names, data types, units, error states and timestamps. The application layer can then integrate one component at a time, beginning with basic successful requests and progressing to missing data, invalid-input and unavailable-service cases. This approach allows integration defects to be isolated without confusing them with model-performance issues.

Source control should be used to maintain a shared history of changes, while branches or equivalent review practices should reduce the risk of conflicting edits. Changes to shared interfaces should be communicated to all affected component owners. Each integration change should be checked with relevant unit tests and a regression test of previously working functions. Issues should record the observed behavior, steps to reproduce it, expected behavior, assigned owner, and resolution. Before a demonstration or submission, the team should identify a known-good version and verify that the setup instructions work in a clean environment.

The project process should be described using only the practices actually followed. If the team used a planning board, meetings, or code reviews, summarize the process and cite the corresponding project evidence. If a practice was not used, do not claim that it was.

Testing should be organized at several levels. Component tests check individual transformations, model calls, and outputs. Interface tests verify that each component

accepts the agreed request format and returns the expected response structure. Integration tests check that the common application can display outputs from all four components. Regression tests confirm that changes do not break previously working functions. End-to-end tests follow representative user tasks, such as checking a vehicle's condition, reviewing a battery-health forecast, planning a charging stop, and asking the assistant for an explanation. Test outcomes and defects should be recorded with dates and the tested software version.

5.3 Deployment and operational configuration

The integrated application should be packaged and configured so that the same documented procedure can be used for demonstration and evaluation. The deployment procedure should specify the required runtime, dependencies, model files, data locations, environment variables, external-service configuration, and startup order. Configuration values that vary between development and demonstration environments should be kept separate from source code. Where model files are loaded at startup, the application should verify that the files exist and match the expected feature schema before accepting prediction requests.

At startup, the system should check that essential components are available and report any optional services that are unavailable. Runtime logs should record relevant errors, component status, model/version information and request timing without unnecessarily storing personal information, voice content or precise location history. If the application depends on live charging stations, route, traffic or language-model services, it should handle network errors and timeouts and make the limitations visible to the user. The system should distinguish a live result from a cached or demonstration result.

Backups and restoration should cover the project artefacts that are needed to reproduce the demonstration: source code, configuration templates, model artefacts, database records where applicable, and the EV knowledge base. Secrets must not be included in ordinary backups or committed to source control without appropriate protection. A recovery procedure should explain how to restore the required files and re-run the application. Monitoring can include application availability, failed

requests, model-loading errors and response times; any thresholds reported as operational results must be measured in the actual environment.

This section describes the deployment controls required for a reproducible integrated prototype. Before submission, replace general descriptions with the exact deployment method actually used, including the target environment and the commands or steps needed to run it. If the system was evaluated only on a development machine, state that clearly rather than implying production deployment.

5.4 Evidence of team execution

Evidence of execution should demonstrate how the four individual components were developed and brought together. Appropriate evidence includes repository commits associated with each component, the final integrated source-code version, documented interface definitions, meeting decisions, task-board records if used, component test reports, integration test results, screenshots of the integrated application, and research or development diaries. These records should be summarized in the report and retained in an appendix or project repository where permitted; raw activity logs should not replace a concise explanation of what the evidence demonstrates.

For this project, the evidence package should map each component to its owner and identify the principal artefacts produced: anomaly-detection code and evaluation records; battery-health forecasting code, model comparison and recommendation rules; charging and range prediction code with its evaluation outputs; and the RAG/LLM and voice-interaction implementation with representative test cases. Shared evidence should include the common application interface, the agreed data contracts, cross-component test cases, issue-resolution records, and a reproducible demonstration procedure. Each item should include enough context to identify the version tested and the person or component responsible.

The component-level evaluation reports the following results [1]: anomaly-detection examples produced a score of 0.08 for normal operation, 0.96 for a charging-system anomaly and 0.91 for a motor/thermal anomaly; the battery-health comparison selected Random Forest over XGBoost across the reported 3-, 6- and 12-month

horizons; the range predictor reported an RMSE of 4.12 km and MAE of 3.05 km; the queue predictor reported an MAE of 2.4 minutes; route optimization reported a performance score of 94.2%; and charging-time/cost estimation reported a performance score of 93.6% with a charging-time variation margin of ±3.5 minutes. These figures should be presented as reported evaluation results unless the team has independently reproduced them in the current integrated implementation.

The virtual-assistant prototype combines EV-specific retrieval, an LLM, and text/voice interaction [1]. No numerical assistant-accuracy or user-acceptance metric is provided here, so the report should avoid assigning an unsupported accuracy percentage to this component. Instead, provide the actual test cases and observations available to the team, such as whether relevant knowledge was retrieved, whether answers were grounded in the knowledge base, whether text and voice input worked, and how the system behaved when speech recognition or retrieval failed.

No specific MS Planner, Teams, Git history, meeting-minute or research-diary records were supplied for this draft. Before final submission, insert verifiable evidence of references for the activities that occurred and remove any evidence category that the team did not use. The team should confirm that the contribution matrix is accurate and that all four members agree with the description of individual and shared work.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The evaluation framework for the Electric Vehicle Management System Using Artificial Intelligence and Machine Learning focuses on assessing the accuracy, reliability, usability, and overall performance of the four integrated individual components. These include Adaptive Battery Health Prediction and Personalized Recommendations, the AI-Based EV Virtual Assistant using RAG with Voice Support, Intelligent EV Charging Optimization and Range Prediction, and AI-Based EV Anomaly Detection. Success criteria include prediction accuracy, relevance of recommendations, accuracy of retrieved information, response time, anomaly detection performance, and the system's ability to operate as a unified solution.

6.2 Integration and end to end testing

Integration and end-to-end testing ensure that all four components function together effectively within the proposed EV management system. Testing focuses on data exchange between components, the consistency of outputs, system responsiveness, and the handling of user queries and EV-related information. The testing process also evaluates whether battery health predictions, charging and range recommendations, anomaly alerts, and virtual assistant responses provide consistent and useful information throughout the user journey.

6.3 Overall results

The overall results are evaluated based on the performance of the four individual components and their contribution to the common objectives of the research project. The system aims to provide intelligent EV management through battery health monitoring, personalized recommendations, charging optimization, range prediction, anomaly detection, and conversational assistance. The combined solution is expected to improve the accessibility of EV information, support informed decision-making, and enhance the efficiency and reliability of EV usage. Actual performance metrics and experimental results should be included based on the completed implementation and testing.

6.4 Stakeholder and user validation

Stakeholder and user validation assess whether the integrated system meets the needs of electric vehicle owners, drivers, and other relevant stakeholders. Feedback can be collected through demonstrations, usability testing, questionnaires, and interviews to evaluate the usefulness, accuracy, ease of use, and practicality of the system. Particular attention is given to the clarity of battery health recommendations, the usefulness of charging and range predictions, the reliability of anomaly alerts, and the effectiveness of the voice-enabled virtual assistant. The feedback can guide further improvements to the system.

6.5 Achievement of common objectives

The research project aims to achieve its common objectives by combining artificial intelligence and machine learning technologies to provide a comprehensive EV management solution. The battery health component supports battery monitoring and personalized recommendations, while the charging optimization component improves charging decisions and range estimation. The anomaly detection component identifies unusual EV-related patterns, and the RAG-based virtual assistant provides context-aware information through text and voice interactions. Together, these components aim to improve EV usability, support preventive maintenance, optimize energy utilization, and provide intelligent assistance to EV users

6.6 Integrated limitations and threats

The integrated system may face limitations related to the availability and quality of EV datasets, prediction accuracy, computational requirements, real-time data access, and compatibility between individual components. Battery health estimation, charging optimization, range prediction, and anomaly detection may be affected by variations in battery conditions, driving behavior, environmental factors, and vehicle models. The RAG-based virtual assistant may also produce incomplete or inaccurate responses if the knowledge base is limited or outdated. In addition, privacy, cybersecurity, scalability, and dependence on external services represent potential threats that must be considered during implementation and future development.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The proposed EV management system offers an integrated AI-powered solution that helps electric vehicle users make informed decisions about battery health, charging, driving range, maintenance, and EV-related information. Its main value proposition is the combination of predictive analytics, anomaly detection, personalized recommendations, and a voice-enabled virtual assistant within one platform. Potential beneficiaries include EV owners, drivers, fleet operators, EV service centers, charging station providers, and vehicle maintenance businesses. By providing timely insights and accessible assistance, the system aims to improve user convenience, reduce avoidable maintenance costs, and support more efficient EV operation.

7.2 Market alternatives and differentiation

Existing EV management solutions may provide features such as battery monitoring, charging management, driving range estimation, vehicle diagnostics, or customer support. However, the proposed system aims to differentiate itself by integrating four complementary AI-based components into a single EV management platform. Its combination of adaptive battery health prediction, intelligent charging and range optimization, anomaly detection, and a RAG-based virtual assistant with voice support provides a more comprehensive approach to EV assistance. The system's potential competitive advantage lies in the integration of these capabilities, personalized guidance, and conversational access to EV-related knowledge.

7.3 Business model and feasibility

The proposed system could be commercialized as a web-based or mobile application using a freemium or subscription-based business model. Basic features could be provided to individual EV users, while premium services such as advanced battery analytics, personalized recommendations, detailed charging optimization, and fleet monitoring could be offered through paid plans. Business-to-business opportunities may include partnerships with EV dealers, charging station operators, fleet

management companies, and maintenance service providers. Commercial feasibility depends on development and infrastructure costs, access to reliable EV data, model performance, user demand, scalability, and the ability to establish suitable industry partnerships.

7.4 Adoption intellectual property and compliance

Successful adoption of the system requires a user-friendly interface, reliable predictions, accurate recommendations, secure data handling, and compatibility with available EV data sources. Demonstrations, pilot testing, user feedback, and partnerships with relevant automotive stakeholders could support wider acceptance. Intellectual property considerations include ownership of the software code, AI models, datasets, system architecture, and original research contributions. The project should also consider applicable data protection and privacy requirements, cybersecurity practices, software licensing, and the usage conditions of third-party AI models and external services. Clear documentation and responsible data management are essential for building trust and supporting future commercialization.

7.5 Sustainable Development Goal contribution

The proposed EV management system can contribute to several United Nations Sustainable Development Goals (SDGs). It is particularly relevant to SDG 7: Affordable and Clean Energy, by supporting efficient charging and energy utilization, and SDG 9: Industry, Innovation and Infrastructure, through the development of an integrated AI-based EV management solution. It also supports SDG 11: Sustainable Cities and Communities, by encouraging the effective use of electric mobility, and SDG 12: Responsible Consumption and Production, by promoting battery health monitoring, preventive maintenance, and potentially longer battery service life. Through these contributions, the project aims to support sustainable transportation and responsible technology innovation.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

This research introduces IntelligentEV, an intelligent Electric Vehicle Management System based on Artificial Intelligence (AI) and Machine Learning (ML) solutions, to solve the issues concerning vehicle diagnostics, battery health management, charging optimization and technical assistance problems. The primary objective was to integrate all these functions into a single mobile decision support system, so as to remove the need to use a plethora of individual applications.

This solution will include four integrated components: AI-Based EV Anomaly Detection, Adaptive Battery Health Prediction and Personalized Recommendations, Intelligent EV Charging Optimization and Range Prediction, and an AI-Based EV Virtual Assistant with Retrieval-Augmented Generation (RAG) and voice support.

Results of the component level assessments were reported to be good. The anomaly detection part could detect whether the operating condition is abnormal, and the battery health prediction part ran both two algorithms: Random Forest and XGBoost and achieved the best results when using the Random Forest algorithm when forecasting for 3 months, 6 months, and 12 months. In the charging-queue, the charging and range component reported an RMSE and an MAE of 4.12 km and 2.4 minutes, respectively. The virtual assistant was used to show the knowledge repository and the conversation with a particular focus on EVs.

The overall project aims at establishing technical feasibility of smart EV functionality integration in a single decision support system. But additional end-to-end testing and field validation is needed to ensure the dependability and operational viability of the overall system.

8.2 Principal integrated contribution

The design and integration of four complementary AI/ML based components into one EV management system is important. All these features can be combined in the proposed system, thus providing a more comprehensive knowledge of the EVs and better decision on their use.

The architecture of the built-in system offers a unified application user interface to access battery health forecasts, personalized recommendations, charging and range data, anomaly alerts and conversational assistance. It is designed to be modular, that is, the individual component can be built without affecting the other components and will be easily integrated and communicated with in the future.

The other notable addition is hybrid predictive analytics and actionable decision-making support. The system will provide technical outputs of the vehicle through an access application, to help EV owners make better charging, driving and vehicle maintenance decisions.

The project also provides a foundation for further research on combined intelligent transportation applications and a practical example of the complementarities of the different AI techniques in the scope of an integrated solution.

8.3 Future directions

In future research, the reliability and scalability of the IntelligentEV system and the system implementation in real applications are the next steps.

The first is that the individual AI & ML models need to be tested with a bigger and more representative set of real-world EV data, containing a variety of metrics related to driving & weather, as well as car models, and battery type. This would help to know the extent of the model's ability to generalize the results and increase the reliability of the predictions.

Second, more complex integration and end-to-end test should also be performed to ensure the performance of the components when the systems are operated under realistic conditions, to communicate, deal with errors, respond consistently and behave according to the intended design.

Third, integrating real-time vehicle data, charging stations, traffic data and electricity tariffs can further enhance the precision and usefulness of monitoring, predicting, and optimizing functions.

An expanded EV knowledge base, multilingual support, and better voice recognition could further enhance the capabilities of the AI-based virtual assistant.

Recommendations may be more dynamic, based on more up-to-date data on the use of vehicles.

Finally, user testing, security enhancements, and optimization for deployment and collaboration with service providers for EVs could be considered for future adoption and commercialization. The improvements will help make IntelligentEV more reliable and practical to use for EV management.

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APPENDICES

Appendix A: Integrated interface specification

This appendix provides additional information regarding the interfaces and data exchange requirements for the proposed IntelligentEV system.

The system comprises four integrated components to deliver the proposed solutions, including AI-based EV anomaly detection, battery health prediction and personalized recommendations, intelligent EV charging and range prediction, and AI-based EV virtual assistant.

The integrated solution will adopt a common application interface to provide the diagnostic, predictive, recommendation, and virtual assistant results from the four components.

Table 7 : Integrated Component Interface Specifications

Component	Input Data	**Expected Output**
AI-Based EV Anomaly Detection	Battery voltage, current, temperature, speed, acceleration, charging status, and power consumption	Anomaly score, category, severity level, vehicle status, and diagnostic guidance
Battery Health Prediction and Personalized Recommendations	Historical battery degradation data and separately supplied charging and driving behavior	SOH predictions for 3, 6, and 12 months and personalized battery care recommendations
Intelligent EV Charging Optimization and Range Prediction	SOC, vehicle details, driving conditions, charging-station information, and route data	Estimated driving range, charging-station waiting time, optimized routes, charging time, and estimated cost
AI-Based EV	User questions, available	Context-aware EV information

Virtual Assistant	vehicle context, and EV knowledge-based information	and guidance through text or voice
Integrated Application	User requests and outputs from the four components	Unified presentation of vehicle diagnostics, predictions, recommendations, and assistance

The information in Table A.1 highlights the logical data input and output requirements for the integrated components.

The interface specifications will enable the common application to deliver the individual functionality in a unified interface for managing EVs.

Table 8 :Integration Validation and Error-Handling Requirements

Validation Scenario	**Expected System Behavior**
Valid component input	Process the request and return the relevant output
Missing battery or vehicle measurements	Identify missing data and return an appropriate message
Invalid input values	Reject invalid values and provide validation feedback
Unavailable charging information	Inform the user that live charging information is unavailable
Failed model or service request	Return an understandable error without interrupting unrelated functions
Unclear virtual assistant query	Request clarification or indicate insufficient information

Table A.2 highlights the key validation and error-handling scenarios applicable to the integrated solution.

The validation scenarios will ensure the reliable operation of the common interface for the integrated functions.

The interface specifications in Table A.1 and Table A.2 provide logical requirements for implementing the integrated solution.

The actual endpoints, request/response formats, and test results should be provided as part of the implementation to meet the specification.

Appendix B Team contribution declaration

This appendix provides additional information regarding the team contributions to the project.

The project entails four interrelated components managed by the four-team members.

The team members contributed individually to the development and evaluation of the proposed solution components while collaborating on the integration and documentation efforts.

Table 9 : Individual Research Team Contributions

Team Member	**Student ID**	**Assigned Component**	**Principal Contribution**
Vidushan A A D D N	IT22157928	AI-Based EV Anomaly Detection System	Development and evaluation of anomaly detection techniques to identify abnormal vehicle operating conditions and generate diagnostic information Development and evaluation of anomaly detection techniques to identify abnormal vehicle operating

			conditions and generate diagnostic information
Fernando C T H H D	IT22247490	Adaptive Battery Health Prediction and Personalized Recommendations	Development and evaluation of Random Forest and XGBoost models for SOH prediction and a rule-based personalized recommendation mechanism
Dias N. T. B. P	IT22134080	Intelligent EV Charging Optimization and Range Prediction System	Development of driving-range prediction, charging queue estimation, route optimization, and charging time and cost estimation functions
Jayarathne W.T.D. K	IT22071170	AI-Based EV Virtual Assistant	Using RAG with Voice Support Development of EV knowledge retrieval, LLM-based response generation, and text and voice interaction capabilities

Shared Contributions

The team members' shared contributions could include the following:

- Defining the research problem, aim, and objectives.
- Conducting a literature search and identifying research gaps.

- Establishing the system's overall architecture and component integration requirements.
- Coordinating the testing and evaluation of all four components.
- Preparing the integrated research report, reviewing the results, and standardizing the terminology.
- Discussing system limitations, ethical implications, and improvements.